T-5 Hypotension/Shock Protocol: Adult and Pediatric
Effective: October 3, 2024
Revised: July 2025

Hypotension can be defined as a systolic blood pressure of less than 90 in the adult and B/P below the 5th percentile in a child/infant.

Infant less than 1 month	60 mmHg systolic
1 month to 1 year	70 mmHg systolic
1 year to 10 years	70 plus 2 X age in years systolic
➢ 10 years	90 mm Hg systolic

This is not always reliable and should be interpreted in context and patient's typical BP if known. Shock may be present with a normal blood pressure initially. Tachycardia may be the only manifestation. In children, early shock may be manifested with poor skin perfusion and altered mental status.

Consider all possible causes of shock and treat per appropriate protocol.

Hypovolemic Shock;

Hemorrhage, trauma, GI bleeding, ruptured aortic aneurysm or pregnancy-related bleeding. burns, dehydration

· Cardiogenic Shock:

- Heart failure: MI, Cardiomyopathy, Myocardial contusion, Ruptured ventricle / septum / valve / toxins.

· Distributive Shock:

- Sepsis (systemic infection)
- Anaphylactic
- Neurogenic: Hallmark is warm, dry, pink skin with normal capillary refill time and typically alert.
- Toxins

· Obstructive Shock:

- Pericardial tamponade; Pulmonary embolus; Tension pneumothorax.

Begin Initial Medical Care

Refer to Multi-system Trauma Care for suspected hemorrhagic shock.

Do not allow a patient in shock to become hypothermic.

Hemorrhagic Shock Adult:

- Utilize the Spinal Motion restriction protocol if appropriate
- Control any major bleeding in accordance with Multiple Trauma protocol
- Tension Pneumothorax requires needle chest decompression in accordance with Multiple Trauma protocol
- Run IVF at KVO if adequate mental status and/or a radial pulse is present in the adult.
- Normal saline 500 ml boluses to maintain mental status if hypotensive or to maintain a radial pulse in the adult if not present.
- Monitor lung sounds. Should crackles develop, decrease IV to KVO and notify Medical Control.
- Consider TXA Administration per protocol (1 Gram mixed in 100ml IV Solution ran over 10min)

Hemorrhagic Shock Pediatric:

- In a pediatric patient not exhibiting signs of pubescence, use Buretrol tubing or a syringe and stopcock device to administer fluid boluses

- In the pediatric patient with normal mental status and peripheral perfusion run IVF at KVO.
- If needed, IVF shall be administered at a rate to maintain adequate mental status or peripheral perfusion. Boluses are to be administered at 20 ml/kg (administer incrementally) and the patient reassessed after each bolus.

- Monitor lung sounds. Should crackles develop, decrease IV to KVO and notify medical Control.
- Consider TXA 15mg/kg over 10min per protocol.

Distributive Shock (Neurogenic shock/Septic shock/Post ROSC shock in Adult:

- Administer Normal Saline or Lactated Ringers 500 ml boluses if B/P < 90 until 90 systolic met.
- If B/P < 90 systolic **WITH signs of poor perfusion** (ie. Mottled skin, decreasing LOC, capnography < 25) administer a 500 ml IV fluid bolus while preparing push dose epinephrine.
- **Push Dose Epinephrine is prepared** by placing 1mg of 1:10,000 or 1:1,000 epinephrine in 100 ml NS. This reconstitutes to a dosage strength of 10 mcg/ml. Label the contents.
- Withdraw 5-10 ml of the drug and administer **1-2 ml** every 3-5 minutes checking B/P prior to administration until a B/P of 90 systolic is obtained. Label the syringe. Once B/P is achieved, place IVF at KVO. Monitor closely and repeat procedures in accordance with protocol.

Anaphylaxis:

- Treat as above except, per anaphylaxis protocol, **administer 1 ml (10mcg) of push dose epinephrine every minute** until symptoms improve.

Distributive Shock (Neurogenic shock/Septic shock/Post ROSC shock in Pediatric:

- In a pediatric patient not exhibiting signs of pubescence, use Buretrol tubing or a syringe and stopcock device to administer fluid boluses
- IVF shall be administered at a rate to maintain adequate peripheral perfusion or B/P at the 5th percentile. Boluses are to be administered at 20 ml/kg (administer incrementally) and the patient reassessed after each bolus.
- Monitor lung sounds frequently. Should crackles develop, decrease IVF to KVO and contact Medical Control.
- Consider push dose Epinephrine.
- Push dose Epinephrine is prepared by
  - Draw the standard cardiac arrest dose (0.01mg/kg) into a 10ml syringe.
  - Draw up enough saline into the same syringe to create a total volume of 10ml.
  - **Administer 1-2ml of solution (0.001-0.002 mg/kg) every 3-5 minutes to the desired effect**

Anaphylaxis refractory to 3 doses of IM epinephrine:

- Treat as above except, per anaphylaxis protocol, place 1 mg of a 1:1 or 1:10 epinephrine into 100ml of NS. This reconstitutes to a dosage strength of 10 mcg/ml.
- **Administer 1 ml (10mcg) of push dose epinephrine every minute** until symptoms improve.

Cardiogenic Shock Adult

- In the hypotensive patient, even if crackles are present, administer one 250ml of normal saline.
- If hypotension is accompanied **WITH** poor perfusion (ie. Mottled skin, decreasing LOC, Capnography < 25) administer a 500 ml fluid bolus if clear lung fields or one 250 ml IV fluid bolus if crackles while preparing **Push Dose Epinephrine.**
- Administer **push dose epinephrine as in distributive shock.**

Cardiogenic Shock Pediatric

- In a pediatric patient not exhibiting signs of pubescence, use Buretrol tubing or a syringe and stopcock device to administer fluid boluses
- Administer 5-10 ml/Kg IVF over 10-20 minutes.
- Repeat if needed IF NO CRACKLES are present.
- Consider push dose Epinephrine.
- Push dose Epinephrine is prepared by
    - Draw the standard cardiac arrest dose (0.01mg/kg) into a 10ml syringe.
    - Draw up enough saline into the same syringe to create a total volume of 10ml.
    - Administer 1-2ml of solution (0.001-0.002 mg/kg) every 3-5 minutes to the desired effect

Hypovolemic Shock Adult Not considered Hemorrhagic:

- If tachycardic and has clear lung fields consider an IVF 500 ml bolus.
- If hypotensive, administer 500 ml IVF boluses to maintain B/P at least 90 systolic.
- If hypotensive and crackles administer 250 ml IVF Bolus and contact Medical Control. If unable to reach Medical Control, administer a second 250 ml IV bolus.

Hypovolemic Shock Pediatric Not considered Hemorrhagic:

- In a pediatric patient not exhibiting signs of pubescence, use Buretrol tubing or a syringe and stopcock device to administer fluid boluses
- In the pediatric patient with normal mental status and peripheral perfusion run IVF at KVO.
- If needed, IVF shall be administered at a rate to maintain adequate mental status or peripheral perfusion.
- Boluses are to be administered at 20 ml/kg (administer incrementally) and the patient reassessed after each bolus.
- Monitor lung sounds. Should crackles develop, decrease IV to KVO and notify medical Control.